

ChE 596 Polymer Rheology - Fall 2024

Homework 5

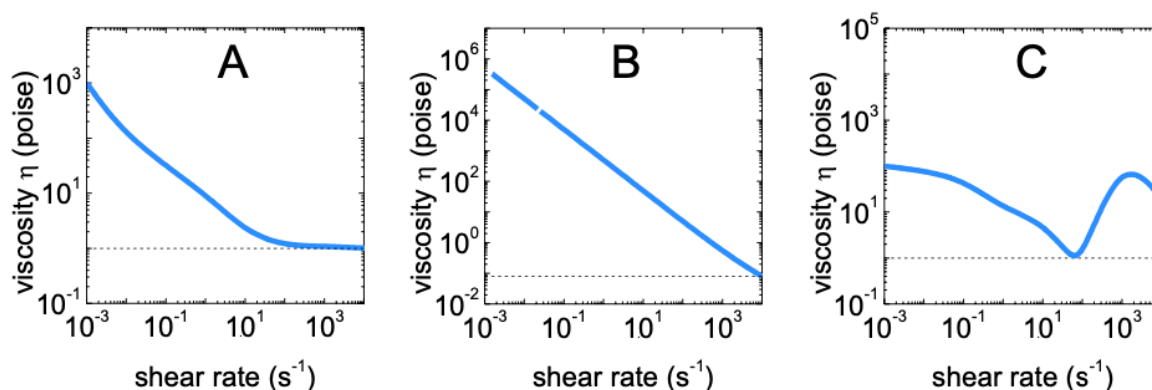
Due date: Tuesday, November 19th at 1:00 pm EDT

Show all your work for full credit.

Problem 1 (20 points)

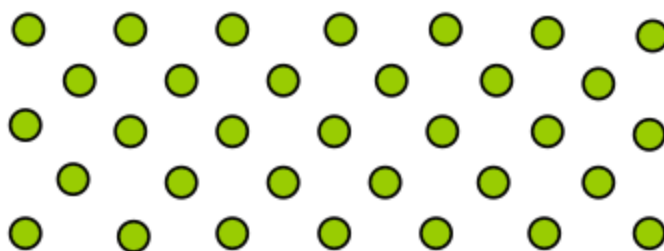
You are asked to formulate a new paint. For this, you start by dispersing an organic pigment (spheres of 0.2 μm diameter) in a non-polar organic liquid of viscosity 0.1 Pa·s (10 Poise).

- Estimate the ratio of the hydrodynamic to Brownian forces in the system at 300K at
 - low shear rates (10^{-4} s^{-1}) (Boltzmann constant $k = 1.38 \times 10^{-23} \text{ J/K}$)
 - high shear rates (100 s^{-1})
- Calculate the diffusion coefficient D of a pigment particle in the organic liquid (m^2/s)
- This pigment is unstable with time (i.e., the particles flocculate and the flocs settle). What would you do to stabilize the particles? Include in your answer what additional component or material you would add to the system and the corresponding mechanism of stabilization.
- Having stabilized the paint, you now need to modify the rheology of the paint for it to be usable. This can be done by addition of rheological modifiers. Given the three possible modifiers below, which would you use and why?



Problem 2 (15 points)

One class of suspensions which we have briefly discussed in class are colloidal crystals, e.g., polystyrene latex spheres ($\sim 0.4 \mu\text{m}$ diameter) in a dialyzed aqueous medium. These systems are called “colloidal crystals” because of the crystal-like ordering of particles within the medium as shown below. The ordering is caused by electrostatic repulsive forces between the particles.



- Sketch the shape of the expected dynamic behavior (G' , G'' versus ω) for the system at small strains (within the LVE) and explain your reasoning.
- Sketch the shape of the viscosity as a function of shear stress for the system and explain your reasoning.
- If the particles were irregular and asymmetric, would you expect to see the ordering behavior? Why or why not?

Problem 3 (15 points)

Winter and coworkers have shown that for a crosslinking system, G'' and G' , when plotted as a function of frequency, become parallel at the gel point. They also determined an expression for the loss tangent at the gel point to be:

$$\text{Loss tangent} = \tan(n\pi/2)$$

where n is the relaxation exponent.

- Explain physically why G'' and G' become parallel at the gel point.
- Determine what values of " n " result in G'' being greater than, equal to, or less than G' at the gel point.
- Richtering et. al studied the physical gelation of a polymer produced by bacteria. Physical gelation occurs by the formation of crystalline domains which act as physical cross-links for the remaining amorphous regions. At a temperature $T=30^\circ\text{C}$, the physical gelation process was monitored using dynamic rheology. The researchers found that the polymer was transformed from a liquid to a gel due to the physical cross-links. Yet, they report the following data i.e., no intersection of the loss tangent. Give two reasons which might explain the behavior below.

